Lab 2 Report

Proportional Control

Name Surname

EEE3094S Control Systems Engineering
STUDENT NO

Mechatronics: Department of Electrical Engineering

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Contents

1 Introduction

The aim of this lab is to determine the specific system model of ... using ... methods. Based on this model, a proportional controller is designed to ...

2 System Modelling

What happens in this section? ... Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

2.1 Methodology

1. First, the list was created.
2. Subsequently, steps were added
3. Lastly, I got tired of writing this method.

2.2 System Model

2.2.1 A free-body diagram illustrating the forces acting on the kitticopter

Explain it

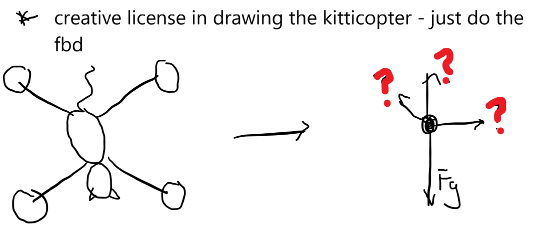

Figure 1: Kam's terrible fbd and copter

2.2.2 A block diagram of the whole system showing the various signals

Explain it

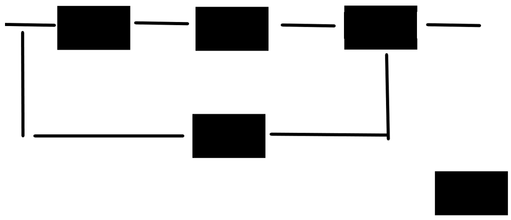

Figure 2: Kam's terrible very nondescript block diagram

2.2.3 A differential equation describing the motion of the mass

Explain it

$$50m3 + terrible = maths - \frac{dy}{yay} \tag{1}$$

2.2.4 A transfer function derived from the differential equation

How'd you do it? What does it represent? From Equation 1 we get...

$$transfer_{f}unctionhere \tag{2}$$

3 System Identification

3.1 Step Response Test

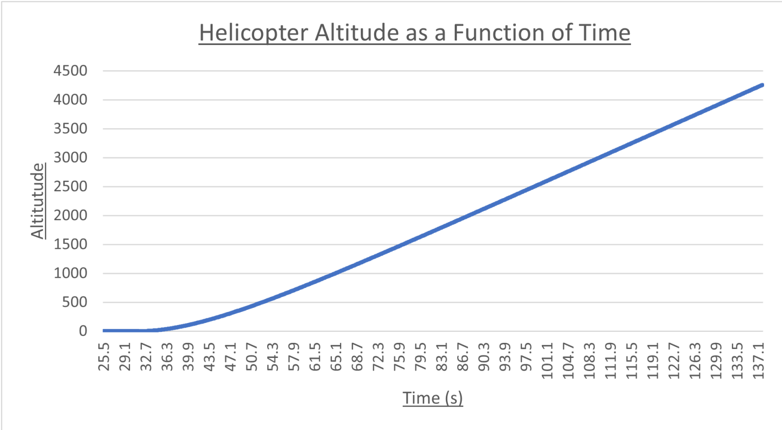

Figure 3: Helicopter Altitude (m) vs Time (s)

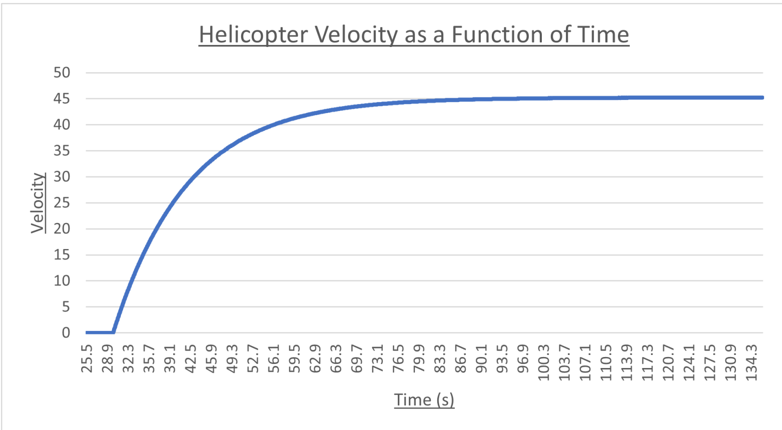

Figure 4: Helicopter Velocity (m/s) vs Time (s): Derivative of Figure 1 (and equal to the transfer response)

Damping Coefficient	**Spring coefficient**
12345	12345

Table 1: System Identification Values from my brain

The transfer function below is obtained...

$$transfer_function here \tag{3}$$

Calculation of the System Identification Values: How did you get the above values? Elaborate more than you would in methodology. Remember to include formulae, etc.

3.2 Model Validation

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How? Why? What were your findings?

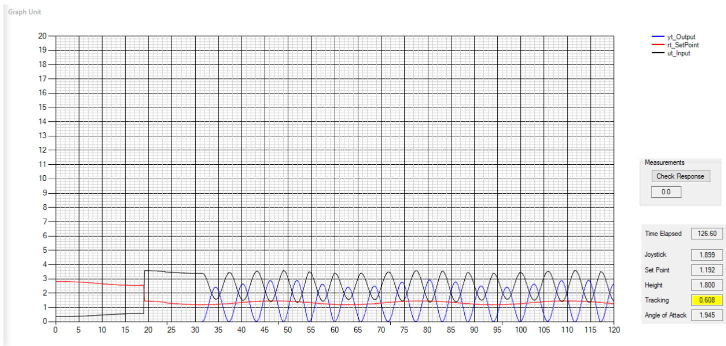

Figure 5: This is a random image, use your own to show some sort of validation or comparison

4 Controller Design

What happens in this section? ...

4.1 Design Process

The proportional controller must achieve these specifications ...

1. Fly.
2. Not crash
3. Work *hopefully*

To determine the controller gain ... Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.
Include description of the analysis techniques used, along with supporting calculations, tables, figures etc.

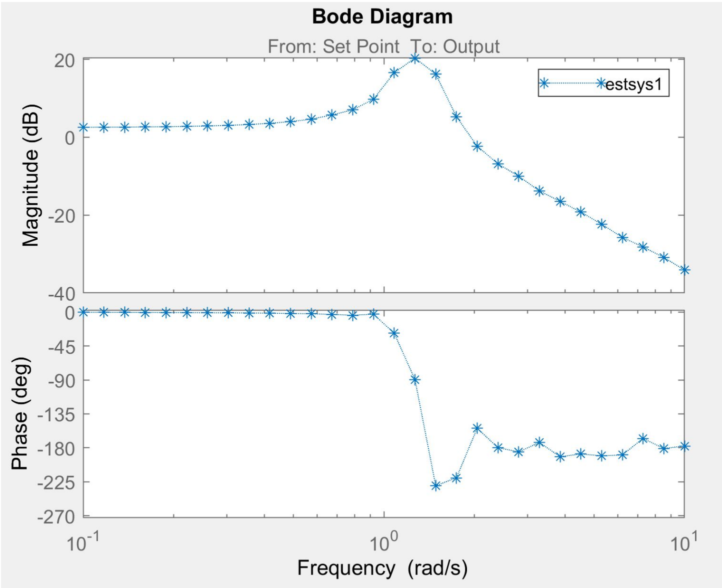

Figure 6: Some kind of time/frequency domain analysis

4.2 Validation

The controller was implemented in SIMULINK, as illustrated in Figure 7.

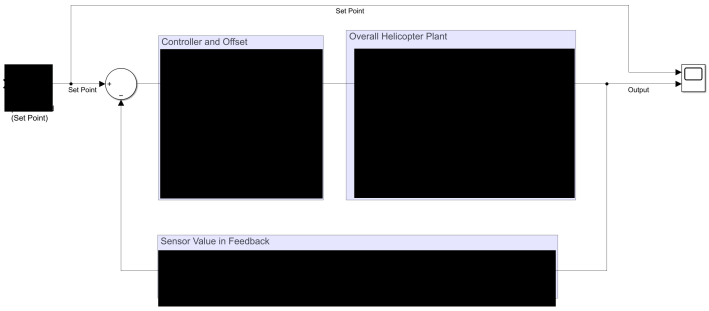

Figure 7: Simulink model

This gave the following results ... Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.
Include figures of your results, what they mean? does your controller meet your specifications? why? why not?

4.3 Circuit Implementation

What type of circuit did you use? Why? What components did you use? What values and why? Include all the details of the physical design of your circuit, a schematic and a photo of it on your Veroboard ... Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

5 Controller Tests

5.1 Tests Performed

How did you test the controller? Include relevant plots and figures ... Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

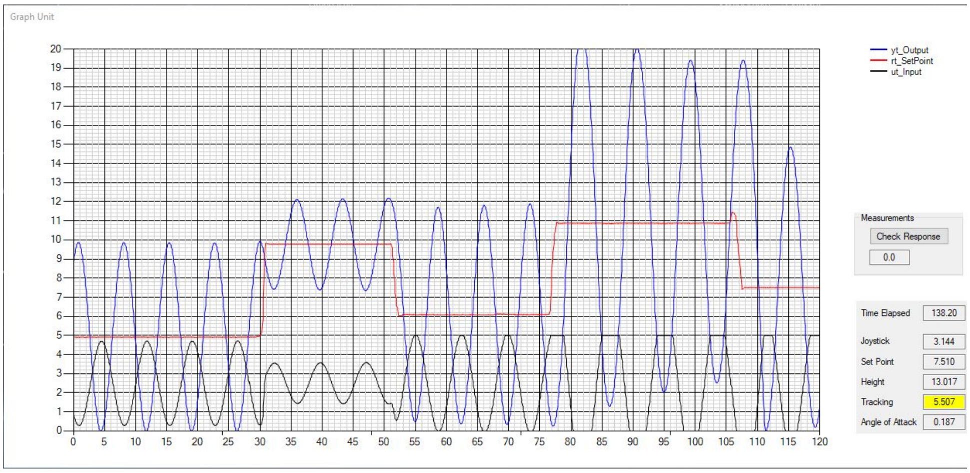

Figure 8: Not you what you want to see

5.2 Performance Evaluation

Evaluate the performance of your controller. Did it meet the specifications? Why and why not? ... Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

6 Conclusion and Recommendations

Rename things/sections as you see fit. Please still refer to the lab manual about what else to include and the rubric for what you'll actually get marks for. Be concise.

Was the aim of the lab achieved? Discuss your results ... Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum..

How would you improve the performance of your controller?
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A Appendix and Notes

Here you can add code or other relevant things that weren't very applicable within the report. Put things in here that you actually reference in the report.

A.1 MATLAB Code

```
% Example code snippet
t = 0:0.01:10;
y = sin(t);
plot(t,y);
```

A.2 Additional Figures

Add extra supporting figures here if needed.